

1. What is the problem of premature evaluation when defining SageMath function?

- (a) Illustrate your explanations with an example we have covered in the course.
- (b) Use the example to show how to fix this premature evaluation problem.

2. Consider $f_k(x_1, x_2, \dots, x_n) = x_k^2 + \sum_{j=1, j \neq k}^n (2j+1)x_j + (2n)x_k - 7$.

Define a function **F** which takes on input a list **X** of variables and an index **k** in the range of **X**. **F(X, k)** returns $f_k(x_1, x_2, \dots, x_n)$. Show that your function works for $n = 8$.

3. Let C be the curve defined by $x^4 + x^3y - y^2 = 0$ and the point $P = \left(1, \frac{1+\sqrt{5}}{2}\right)$ on C .

(a) Compute the tangent line of C at P .

(b) Viewing y in function of x , what is $y'(1) = \left. \frac{dy}{dx} \right|_{x=1}$?

4. The arc length of the curve $y = \exp(x)$ is defined by the function $L(t) = \int_0^t \sqrt{1 + \exp(x)^2} dx$. Define the function $L(t)$. What is $L(1)$? Compute $L'(1)$.

5. Write an efficient recursive function to compute

$$q_0(x) = 1, \quad q_1(x) = 1 - x, \quad \text{for } n > 1 : q_n(x) = (3x + 1)q_{n-1}(x) + 2xq_{n-2}(x).$$

Make sure your function can compute $q_{100}(x)$. Do **NOT** use `cached_function`.

6. Let $p = x(y^2 - xz)^2 - y^5 = 0$ and $q = y^4 + y^3z - x^2z^2 = 0$ define two surfaces, respectively called P and Q .

- (a) Plot the two surfaces P and Q , using the colors red and green, with a suitable range for x , y , and z so the curve of intersection $P \cap Q$ is visible.
- (b) Interpret the plot. What can you state about the curve $P \cap Q$?

7. Compute a symbolic-numeric factorization of $p = x^4 - 2x^2 + 999999/1000000$.

What is the accuracy of the factors and the corresponding roots?

Let q be the expanded polynomial obtained by multiplying the computed linear factors. Compare the coefficients of q with the coefficients of p .

8. Find the points on the sphere $x^2 + y^2 + z^2 - 4 = 0$ closest to the point $P = (3, 1, -1)$.

- (a) Using Lagrange multipliers, set up a polynomial system.
- (b) Convert this polynomial into triangular form.
- (c) What are the closest points to P on the sphere?

9. Set up the following linear programming problem: maximize $f(x_1, x_2) = 8x_1 + 11x_2$, subject to $x_1 \geq 0$, $x_2 \geq 0$, $5x_1 + 4x_2 \leq 40$, $-x_1 + 3x_2 \leq 12$.

Solve the problem, report the optimal value of f and the corresponding values for x_1 and x_2 .